



CCODEL
RAISING A NEW GENERATION OF
GLOBAL LEADERS

Computer Programming I (Java 1)

COS221

Mr. Otavie Okuoyo

Phd (in view Computer Science)

College of Science and Technology



Visit Our Website

ccodel.covenantuniversity.edu.ng



MODULE TWO: JAVA SETUP

UNIT ONE: INTRODUCTION TO JAVA AND JAVA DEVELOPMENT ENVIRONMENT

INTRODUCTION TO JAVA AND JAVA DEVELOPMENT ENVIRONMENT

Learning Objectives

By the end of this unit, you will be able to:

- **explain the fundamental concepts of Java programming.**
- **demonstrate how to install and configure the Java Development Kit (JDK) on various operating systems.**
- **configure the environment variables required to run Java applications.**

Learning Outcomes

At the end of this unit, you should be able to:

- **understand the fundamental concept of Java programming.**
- **install and configure the Java Development Kit (JDK) on different operating systems.**
- **set up environment variable necessary for running Java applications.**

Introduction

- **Java is a versatile, Object-Oriented Programming language used to build scalable applications across multiple platforms. Since its launch in 1995, it has remained a top choice for web, desktop, enterprise, and mobile development.**
- **Its "Write Once, Run Anywhere" (WORA) capability, powered by the Java Virtual Machine (JVM), ensures cross-platform compatibility.**
- **This unit introduces Java programming, covering key concepts, syntax, and development tools. Learners will explore the installation and configuration of the Java Development Kit (JDK).**

Getting Started

- **To get started in Java programming, you have to download and install the Java Development Kit (JDK).**
- **The JDK is a software development environment used for developing Java applications. It provides the tools needed to write, compile and run Java application.**

JDK Installation

To download and install the JDK, visit the URL and choose the Operating System (OS) of your choice:
<https://www.oracle.com/ng/java/technologies/downloads/>

JDK 23 JDK 21 GraalVM for JDK 23 GraalVM for JDK 21

Java SE Development Kit 23.0.2 downloads

JDK 23 binaries are free to use in production and free to redistribute, at no cost, under the [Oracle No-Fee Terms and Conditions](#) (NFTC).

JDK 23 will receive updates under these terms, until March 2025, when it will be superseded by JDK 24.

Linux macOS **Windows**

Product/file description	File size	Download
x64 Compressed Archive	228.77 MB	https://download.oracle.com/java/23/latest/jdk-23_windows-x64_bin.zip (sha256)
x64 Installer	205.27 MB	https://download.oracle.com/java/23/latest/jdk-23_windows-x64_bin.exe (sha256)
x64 MSI Installer	204.02 MB	https://download.oracle.com/java/23/latest/jdk-23_windows-x64_bin.msi (sha256)

Environment Variables Configuration

To configure the environment variable, follow the steps:

- 1. Locate the JDK Installation Path: Open your file explorer and navigate to the directory where the JDK is installed. By default, it is installed at the location:**

C:\Program Files\Java\jdk-<version>\bin

Environment Variables Configuration

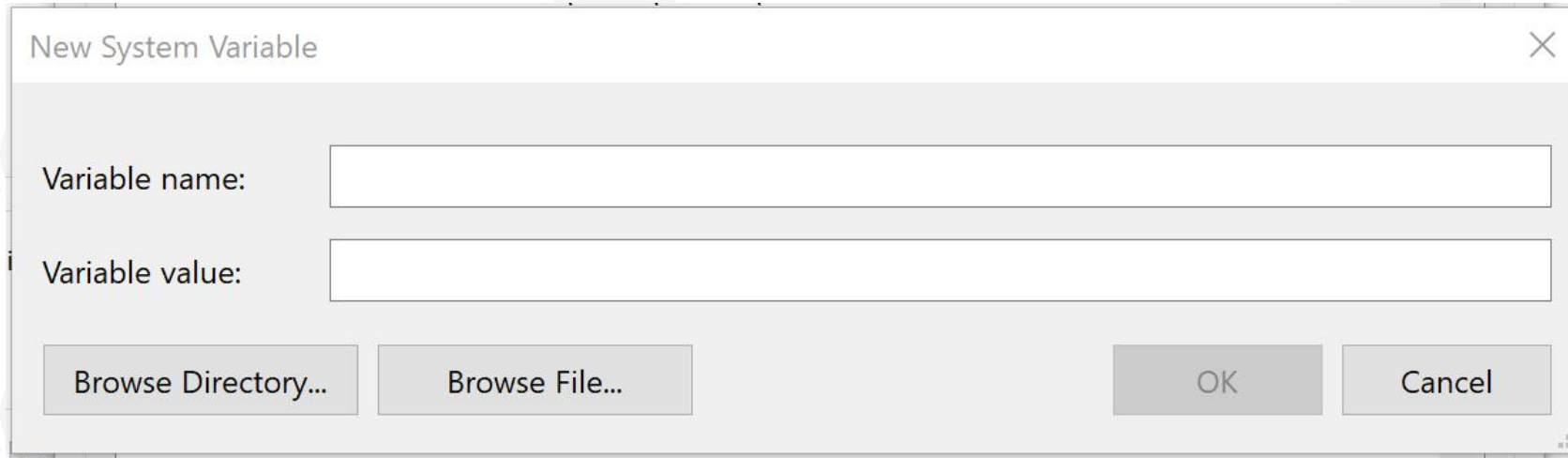
2. **Set the Environment Variable: Right-click on "This PC" > click on "Properties" > click on "Advanced system settings" > "Environment Variables" under System variables click "New". See the in the Figure:**

Variable name: JAVA_HOME

Variable value: C:\Program Files\Java\jdk-<version>

3. **Click on the OK button.**

Environment Variables Configuration



New System Variable

Variable name:

Variable value:

- 4. Verify your installation: After the installation, verify your installation on the Command Prompt using the command below:**

java --version

JDK Installation Verification

For Windows users, to check if your PC has Java installed, open the Command Prompt and type the command below:

```
> java --version
```

If java is installed, you get something similar to what is shown below:

java 23 2024-09-17

Java(TM) SE Runtime Environment (build 23+37-2369)

*Java HotSpot(TM) 64-Bit Server VM (build 23+37-2369,
mixed mode, sharing)*

In-Text Questions (ITQs)

- 1. What does "WORA" stand for in Java development?**
- 2. What are some common areas where Java is used?**
- 3. What is the role of the Java Virtual Machine (JVM)?**
- 4. What is the Java Development Kit (JDK)?**

In-Text Answers (ITAs)

- 1. "WORA" stands for "Write Once, Run Anywhere", referring to Java's ability to run the same code on multiple platforms using the Java Virtual Machine (JVM).**
- 2. Java is widely used in web, desktop, enterprise, and mobile application development.**
- 3. The JVM enables Java programs to run on any platform, ensuring cross-platform compatibility.**
- 4. The JDK is a software development environment used to develop Java applications.**

Conclusion

- **Proper installation and configuration of the JDK and environment variables are crucial for Java development.**
- **By following these setup steps, learners ensure that their systems are ready to compile and run Java applications efficiently, laying a strong foundation for further learning in Java programming.**

Summary

- **This unit introduces Java as a versatile, platform-independent programming language widely used in various application domains.**
- **It guides learners through the essential first steps of setting up the Java Development Kit (JDK), including downloading and installing it on different operating systems.**
- **The unit also explains how to configure environment variables, such as JAVA_HOME, to enable easy compilation and execution of Java programs from the command line.**